

Unitary Coupled Cluster theory on Quantum Computers with Parametric Matrix Models

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We present a quantum computing implementation of the nuclear pairing Hamiltonian using the unitary coupled cluster method with double excitations. The Hamiltonian is written in second quantization and mapped to qubit operators via the Jordan–Wigner transformation. Explicit Pauli-string decompositions and quantum circuits are derived for a minimal four-level pairing model. The exponential excitation operators are implemented using Suzuki–Trotter decompositions. Finally, we demonstrate how parametric matrix models can reduce the dimensionality of the Trotter expansion and significantly decrease quantum circuit depth.

I. INTRODUCTION

Quantum computing offers new possibilities for solving strongly correlated quantum many-body systems. Nuclear systems provide an ideal testbed because pairing correlations dominate the low-energy structure of many nuclei.

In this work we study the nuclear pairing Hamiltonian and construct a quantum algorithm based on the unitary coupled cluster doubles (UCCD) ansatz. We show how parametric matrix models (PMMs) can reduce circuit complexity.

$$S_p^+ = P_p^\dagger, \quad (4)$$

$$S_p^- = P_p, \quad (5)$$

$$S_p^z = \frac{1}{2}(N_p - 1), \quad (6)$$

we obtain the commutation relations

$$[S_p^+, S_q^-] = 2\delta_{pq}S_p^z. \quad (7)$$

The Hamiltonian becomes

II. NUCLEAR PAIRING HAMILTONIAN

The pairing Hamiltonian is

$$H = \sum_p \epsilon_p a_p^\dagger a_p - G \sum_{pq} a_p^\dagger a_{\bar{p}}^\dagger a_{\bar{q}} a_q. \quad (1)$$

Introducing pair operators

$$P_p^\dagger = a_p^\dagger a_{\bar{p}}^\dagger, \quad P_p = a_{\bar{p}} a_p, \quad (2)$$

the Hamiltonian becomes

$$H = \sum_p \epsilon_p N_p - G \sum_{pq} P_p^\dagger P_q. \quad (3)$$

III. QUASISPIN SU(2) SOLUTION

The pairing Hamiltonian possesses an SU(2) quasispin symmetry. Defining operators

$$H = \sum_p 2\epsilon_p S_p^z - G \sum_{pq} S_p^+ S_q^-. \quad (8)$$

For the minimal four-level model the Hamiltonian can be diagonalized analytically within the quasispin basis.

IV. COUPLED CLUSTER DOUBLES

The coupled cluster wave function is

$$|\Psi_{CC}\rangle = e^T |\Phi_0\rangle, \quad (9)$$

with doubles excitations

$$T_2 = \frac{1}{4} \sum_{abij} t_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i. \quad (10)$$

V. UNITARY COUPLED CLUSTER

For quantum circuits we use

$$|\Psi_{UCC}\rangle = e^{T-T^\dagger}|\Phi_0\rangle. \quad (11)$$

Restricting to doubles

$$\tau = T_2 - T_2^\dagger. \quad (12)$$

VI. JORDAN–WIGNER MAPPING

Fermionic operators are mapped to qubits using

$$a_p^\dagger = \frac{1}{2}(X_p - iY_p) \prod_{j < p} Z_j, \quad (13)$$

$$a_p = \frac{1}{2}(X_p + iY_p) \prod_{j < p} Z_j. \quad (14)$$

This converts the Hamiltonian into a sum of Pauli strings

$$H = \sum_i h_i P_i. \quad (15)$$

VII. FOUR-LEVEL PAIRING MODEL

Consider four single-particle orbitals. The pair excitation operator is

$$a_3^\dagger a_4^\dagger a_2 a_1. \quad (16)$$

After Jordan–Wigner transformation we obtain Pauli operators such as

$$X_1 X_2 Y_3 Y_4 - Y_1 Y_2 X_3 X_4. \quad (17)$$

VIII. QUANTUM CIRCUIT

The exponential of a Pauli string is implemented using CNOT ladders and a single-qubit rotation

IX. PARAMETRIC MATRIX MODELS

Instead of assigning independent amplitudes to all double excitations, we write

$$T_2(\theta) = \sum_k \theta_k M_k. \quad (18)$$

This reduces the parameter count and circuit depth.

X. BENCHMARK: UCCD VS PMM

The number of double excitations scales as

$$N_{UCCD} \sim N^4. \quad (19)$$

Parametric matrix models reduce this to

$$N_{PMM} \sim N^2. \quad (20)$$

Consequently the number of Trotter steps and entangling gates is drastically reduced.

XI. EXAMPLE QUANTUM IMPLEMENTATION

Below is a minimal Qiskit implementation of the pairing Hamiltonian.

```
from qiskit import QuantumCircuit
from qiskit.circuit import Parameter
```

```
theta = Parameter('theta')
```

```
qc = QuantumCircuit(4)
```

```
qc.cx(0,1)
qc.cx(1,2)
qc.cx(2,3)
```

```
qc.rz(theta,3)
```

```
qc.cx(2,3)
qc.cx(1,2)
qc.cx(0,1)
```

```
print(qc)
```

This circuit implements the exponential of the Pauli string generated by the UCCD excitation operator.

XII. CONCLUSIONS

We have presented a quantum computing framework for the nuclear pairing Hamiltonian based on the unitary coupled cluster doubles method. Explicit Jordan–Wigner mappings, quantum circuits, and Pauli-string decompositions were derived.

Parametric matrix models offer a powerful method for reducing circuit depth and enabling simulations of larger nuclear systems on near-term quantum devices.